

4-2-1: Examining Wireless Network Issues

At the end of this episode, I will be able to:

1. Identify and explain the significance of wireless network issues, given a scenario.

Learner Objective: *Identify and explain the significance of wireless network issues, given a scenario.*

Description: In this episode, the learner will examine common issues facing wireless networks. We will examine interference, signal degradation and loss, client disassociation, roaming issues, and more.

- What are some common wireless issues, such as interference?
 - o The disruption of wireless communication by external signals, causing:
 - ♣ Reduced signal quality
 - ♣ Decreased network performance,
 - ♣ Connection losses
- Is this the same with channel overlap?
 - o Specific interference type, caused by multiple access points using overlapping channels, causing (common in 2.4 GHz band):
 - ♣ Network congestion
 - ♣ Degraded performance
- What is a client disassociation issue?
 - o When a wireless client loses its connected status with a wireless access point, severing the wireless link.
- What can cause this in wireless networks?
 - o Low signal strength
 - ♣ Weak Wi-Fi signals, causing unreliable connections, frequent disconnections
 - o Network Congestion
 - ♣ High traffic volumes, causing packet loss, disassociation.
 - o Interference
 - ♣ Overlapping frequencies from devices, causing, communication issues
 - o Misconfigurations
 - ♣ Incorrect settings, preventing stable connections.
 - o Outdated firmware
 - ♣ Old software can lead to compatibility and performance issues.
 - o Physical obstructions
 - ♣ Walls and objects attenuate or degrade signals, weakening connectivity.
 - ♣ Network congestion
 - ♣ Degraded performance
- What is roaming,
 - o An automatic process in which a wireless device moves between access points
 - o Done to maintain a stable network connection as a wireless device's location changes
- What are some issues with roaming that we can encounter?
 - o Threshold Settings:
 - ♣ Incorrect signal strength thresholds can cause premature or delayed roaming, affecting network performance.
 - o Overlap Between APs:
 - ♣ Lack of sufficient coverage overlap can create dead zones, disrupting seamless roaming.
 - o Load Balancing Configurations:
 - ♣ Improper client load balancing across APs can force devices to suboptimal roaming,

reducing efficiency.

- o Compatibility Issues:

- ♠ Mismatches in roaming protocols between devices and APs can lead to inefficient roaming behaviors.

- o SSID Configurations:

- ♠ Multiple SSIDs without proper management may cause frequent and unnecessary scanning, impacting device performance.

- Additional Resources:

- o If applicable